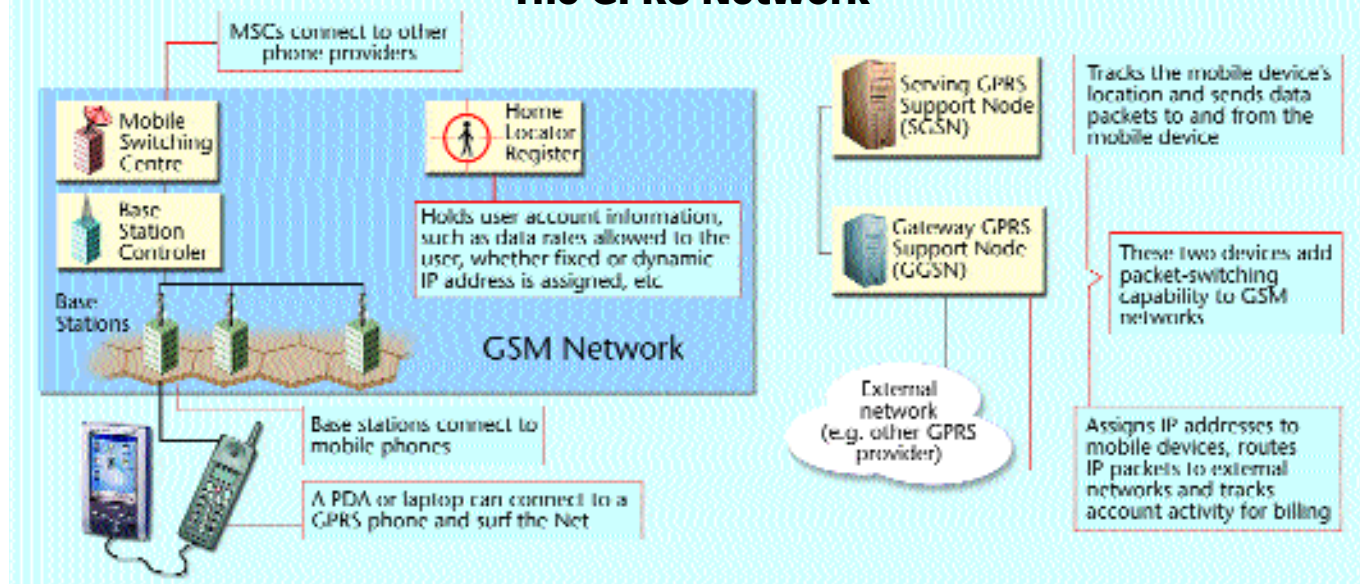


The GPRS Network



channels, with each channel divided into eight slots, each of which can support a throughput of 13 Kbps. Theoretically, a user can use all eight slots simultaneously, but service providers are likely to limit the number of download slots to four or less, and upload slots to one, to allow more concurrent users.

Putting it to use

Surfing on a cell phone may not be fun, given the display area, but you can use it to check your e-mail, news and even sports scores. For heavy-duty tasks, you can connect the phone to a handheld or a laptop and use it as a modem, giving you complete access to the Net.

With more bandwidth, broadband applications such as Voice over IP (VoIP) and videoconferencing will be possible. As the service becomes inexpensive, these options will become more viable and may

even save on communication costs—you can use an IM or VoIP instead of STD and ISD. GPRS phones already allow you to send SMS messages over the Internet instead of the GSM network, thus circumventing the charge for the SMS.

GPRS tools

GPRS devices are of three kinds: Class A devices can handle voice and data simultaneously—you can receive a voice call even while IP data is being transferred. Class B devices can handle voice and data sequentially. The network will notify you of an incoming call, but you will have to suspend the data session to accept it. Class C devices can handle only data.

To use GPRS, you need a GPRS-enabled phone that can be connected to a laptop or a handheld and used as a modem as well as a browser. Some of the GPRS-enabled phones available in India are:

Sony Ericsson T68i: This slick phone has an impressive colour display. It supports GPRS, Bluetooth, High-speed Circuit-Switched Data (HSCSD), Triple-band GSM (900/1800/1900), Enhanced Messaging Service (EMS) and WAP 2.0. SyncML synchronises the online calendar and contacts, and the Wireless Transport Layer Security (WTLS) class 3 provides security during mobile commerce transactions.

Siemens S45: It has 360 KB of onboard memory and its internal data formats are compatible with traditional PC formats such as wave, MIDI, text, etc. The S45's

high-resolution screen, though monochrome, can accommodate up to seven lines of text and is a boon for browsing on the phone.

Nokia 8310: This is Nokia's first GPRS model, but the five-line display may not be the best for surfing. It also has an FM receiver, but supports only mono audio.

Philips Fisio 820: The colour display makes for better browsing—the phone supports GSM 900 and 1800, GPRS, WAP and Bluetooth.

Footnotes

Wireless technologies are fast developing to keep up with the ever-growing needs of mobile users. GPRS can currently give theoretical rates of up to 114 Kbps and this is being pushed to 160 Kbps. EDGE offers far better performance, with a maximum theoretical throughput of 384 Kbps.

The next generation of GPRS, Wide-band CDMA, also referred to as the Universal Mobile Telephone System (UMTS), is one of the standards approved as a 3G cellular technology. It provides data rates of 144 Kbps when mobile, 384 Kbps outdoors at pedestrian speeds and 2 Mbps indoors. It uses an entirely new radio technology based on 5 MHz channels, but retains the GPRS architecture and uses the SGSN and GGSN nodes.

The GPRS infrastructure can actually become the core network for 3G systems, supporting a variety of multimedia services including voice, text, digital images and video on mobile devices.

VEER KOTHARI

Security Concerns

Security over GPRS is at various levels. The user must have a Subscriber Identification Module (SIM) card to use the service. The network can request a password using common password protocols. GPRS also encrypts the link. Over external networks, carriers can optionally employ the IPSec protocol. Finally, since communication is based on IP, any security architecture used over IP, such as Virtual Private Networks (VPN), can also be used with GPRS.